Using dynamic lie algebra for detecting effective Hamitonian

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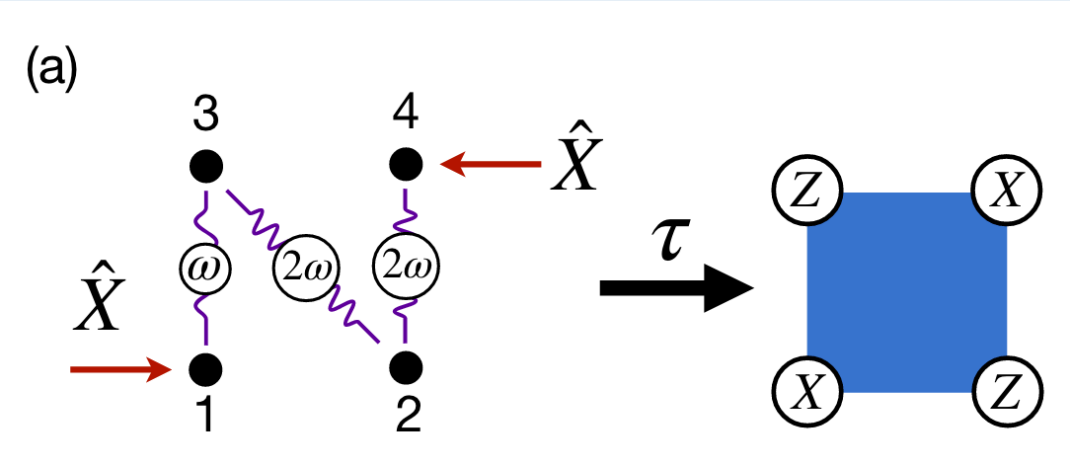
background

Hamitonians of driven/target system

- Driven system: $\hat{H}(t) = \Omega_1 \hat{X}_1 + \Omega_4 \hat{X}_4 + g_{13}(t)\left(\hat{X}_1\hat{X}_3 + \hat{Y}_1\hat{Y}_3\right) + g_{23}(t)\left(\hat{X}_2\hat{X}_3 + \hat{Y}_2\hat{Y}_3\right) + g_{24}(t)\left(\hat{X}_2\hat{X}_4 + \hat{Y}_2\hat{Y}_4\right)$



- Target system: $\hat{H}_\omega = -J\sum_{i,j}\left(\hat{X}_{i,j}\hat{Z}_{i,j+1}\hat{Z}_{i+1,j}\hat{X}_{i+1,j+1}\right)$

background

Possible nested commutators of control operators

- All operators in effective Hamiltonian $H_{i,j}$:

$$\exp\left(-i\hat{H}_{i,j}\tau\right) = \mathrm{T}\exp\left(-i\int_0^\tau \mathrm{d}t\hat{H}(t)\right)$$

Direct solving of evolution operator containning complicated time-sequential intergral(Dyson series).

The Lie-algebra decoupling theorem

Lie-algebra decoupling theorem

- For Hermitian time-dependent Hamitonian $\hat{H}(t)$, the time-evolution operator is:

$$\hat{U}(t) = \mathrm{T}\exp\left(-i\int_0^\tau \mathrm{d}t\hat{H}(t)\right)$$

and

$$H(t) = \sum_{j=1}^{m} G_j(t)\hat{H}_j$$

The Lie-algebra decoupling theorem

It can be extended to larger set with $n \geq m$ elements by taking the commutator of the elements in $\left\{\hat{H}_j\right\}$. The full set of elements $\{\hat{H}_1, ..., \hat{H}_n\}$ form a lie algebra L under commutation of dimision n.

That is, the hamiltonian $\hat{H}(t)$ generates a lie algebra and we can find it by considering all the elements in it:

$$U(t) = \exp(-i * H_{\text{eff}} t)$$

and

$$H_{\text{eff}} = \sum_{j=1}^{n} c_j \hat{H}_j$$

The Lie-algebra decoupling theorem

# commutators	Operators	
0	$\hat{X}_1,\ \hat{X}_4,\ \hat{X}_1\hat{X}_2+\hat{Y}_1\hat{Y}_2,$ $\hat{X}_2\hat{X}_3+\hat{Y}_2\hat{Y}_3,\ \hat{X}_3\hat{X}_4+\hat{Y}_3\hat{Y}_4$	control operators A_n
1	$\hat{Z}_1\hat{Y}_2,\ \hat{Y}_3\hat{Z}_4,\ \hat{X}_1\hat{Z}_2\hat{Y}_3-\hat{Y}_1\hat{Z}_2\hat{X}_3,$ $\hat{X}_2\hat{Z}_3\hat{Y}_4-\hat{Y}_2\hat{Z}_3\hat{X}_4$	commutators $[A_i, A_j]$
2	$\hat{Y}_1\hat{Y}_2,\ \hat{Y}_3\hat{Y}_4,\ \hat{Z}_1\hat{Z}_2\hat{X}_3,\ \hat{X}_2\hat{Z}_3\hat{Z}_4,$ $\hat{X}_1\hat{Z}_2\hat{Z}_3\hat{X}_4,\ \hat{Y}_1\hat{Z}_2\hat{Z}_3\hat{Y}_4,$	commutators $[A_k [A_i, A_j]]$
3	$\hat{Z}_1\hat{Z}_2\hat{Z}_3\hat{Y}_4,\ \hat{Y}_1\hat{Z}_2\hat{Z}_3\hat{Z}_4$	...
4	$\hat{Z}_1\hat{Z}_2\hat{Z}_3\hat{Z}_4$	...

Dynamical Lie Algebra

dynamical Lie algebras

Dynamical Lie Algebras

A different way of looking at this is taking a set of generators $\{iG_j\}$ and asking what kind of unitary evolutions they can generate. This naturally introduces the so-called Dynamical Lie Algebra (DLA), originally proposed in quantum optimal control theory and recently re-emerging in the quantum computing literature. The DLA $i\mathfrak{g}$ is given by all possible nested commutators between the generators $\{iG_j\}$, until no new and linearly independent skew-Hermitian operator is generated. This is called the Lie-closure and is written like

$$i\mathfrak{g} = \langle iG_1, iG_2, iG_3, .. \rangle_{\text{Lie}}.$$

Dynamical Lie Algebra

- Because of the time-symmertric driven Hamiltonian, not all Lie algebra contributes to the effective Hamiltonian.

Dynamical Lie Algebra

Lie - algebra simplify:

- Cartan decomposition: 将 dynamic Lie algebra 拆为两个子空间.

子空间 $h \rightarrow [h, h] \subseteq p$

子空间 $p \rightarrow [p, p] \subseteq h$, $[h, p] \subseteq p$

∴ 分解 g: $g = h \oplus p$

$$\mathfrak{h} = \mathrm{span}\{\hat{Z}_1\hat{Y}_2,\ \hat{Y}_3\hat{Z}_4,\ \hat{X}_1\hat{Z}_2\hat{Y}_3,\ \hat{Y}_1\hat{Z}_2\hat{X}_3,\ \hat{X}_2\hat{Z}_3\hat{Y}_4,\ \hat{Y}_2\hat{Z}_3\hat{X}_4,\ \hat{Z}_1\hat{Z}_2\hat{Z}_3\hat{Y}_4,\ \hat{Y}_1\hat{Z}_2\hat{Z}_3\hat{Z}_4\}\ ,$$

$$\mathfrak{p} = \mathrm{span}\{\hat{X}_1,\ \hat{X}_4,\ \hat{X}_1\hat{X}_2,\ \hat{X}_2\hat{X}_3,\ \hat{X}_3\hat{X}_4,\ \hat{Z}_1\hat{Z}_2\hat{Z}_3\hat{Z}_4,\ \hat{Y}_1\hat{Y}_2,\ \hat{Y}_3\hat{Y}_4,\ \hat{Z}_1\hat{Z}_2\hat{X}_3,\ \hat{X}_2\hat{Z}_3\hat{Z}_4,\ \hat{X}_1\hat{Z}_2\hat{Z}_3\hat{X}_4,\ \hat{Y}_1\hat{Z}_2\hat{Z}_3\hat{Y}_4\}\ .$$

Express $H(t)$ via Magnus Expansion:

$U(t) = \exp(\hat{M}(t))$

$\hat{M}(t) = \sum_{n=1}^{\infty} \hat{M}_n(t)$

$p \rightarrow \hat{M}_1(t) = -i\int_0^t dt'\, H(t')$

$h \rightarrow \hat{M}_2(t) = (-i)^2/2 \cdot \int_0^t dt_1 \int_0^t dt_2\, [\hat{H}(t_1), \hat{H}(t_2)]$

$p \rightarrow \hat{M}_3(t) = -\frac{1}{6}\int_0^t dt_1 \int_0^t dt_2 \int_0^t dt_3\, [\hat{H}(t_1), [\hat{H}(t_2), \hat{H}(t_3)]]$

...

$\hat{M}_{2n}(t) \in h$, $M_{2n+1}(t) \in p$.

choose time-symmetric function

让它消失

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Dynamical Lie Algebra

# commutators	Operators	
0	$\hat{X}_1,\ \hat{X}_4,\ \hat{X}_1\hat{X}_2+\hat{Y}_1\hat{Y}_2,$ $\hat{X}_2\hat{X}_3+\hat{Y}_2\hat{Y}_3,\ \hat{X}_3\hat{X}_4+\hat{Y}_3\hat{Y}_4$	control operators A_n
1	$\hat{Z}_1\hat{Y}_2,\ \hat{Y}_3\hat{Z}_4,\ \hat{X}_1\hat{Z}_2\hat{Y}_3-\hat{Y}_1\hat{Z}_2\hat{X}_3,$ $\hat{X}_2\hat{Z}_3\hat{Y}_4-\hat{Y}_2\hat{Z}_3\hat{X}_4$	commutators $[A_i, A_j]$
2	$\hat{Y}_1\hat{Y}_2,\ \hat{Y}_3\hat{Y}_4,\ \hat{Z}_1\hat{Z}_2\hat{X}_3,\ \hat{X}_2\hat{Z}_3\hat{Z}_4,$ $\hat{X}_1\hat{Z}_2\hat{Z}_3\hat{X}_4,\ \hat{Y}_1\hat{Z}_2\hat{Z}_3\hat{Y}_4,$	commutators $[A_k [A_i, A_j]]$
3	$\hat{Z}_1\hat{Z}_2\hat{Z}_3\hat{Y}_4,\ \hat{Y}_1\hat{Z}_2\hat{Z}_3\hat{Z}_4$	...
4	$\hat{Z}_1\hat{Z}_2\hat{Z}_3\hat{Z}_4$	...

Magnus expansion

Magnus Expansion:

$$\mathcal{T}\exp\left(-i\int_0^T H(t)\,dt\right) \equiv \exp(-iH_{eff}T) \qquad H(t) = H(t+T)$$

$M(t) = \sum_{i=1}^{n} M_n(t)$ 展开为无穷级数

使 $\exp(-iH_{eff}t) = \exp(-iM\cdot t)$

比如: $M = M_1 + M_2 + M_3 + \cdots$

当驱动频率 $\omega = 2\pi/T \gg$ 系统能量尺度, 可压制高阶项.

选择特定驱动波形, 逼近目标哈密顿量.

Magnus expansion

计算过程: Fourier series 表示 $H(t)$

$$H_m = \frac{1}{\tau}\int_0^{\tau} e^{im\omega t} H(t) \qquad H(t) = \sum_{n=-\infty}^{+\infty} H_n e^{in\omega t}, \quad \omega = \frac{2\pi}{\tau}$$

$$[H(t_1), H(t_2)] = \sum_{n,m} [H_n, H_m] e^{i\omega(nt_1 + mt_2)}$$

$$\underset{\tau}{\overset{t_1, t_2}{\Downarrow}} \quad = \sum_{n,m} [H_n, H_m] e^{i\omega\tau(m+n)}$$

if $m+n \neq 0$, high-frequency, ignore.

$\therefore$ We need to preserve ($m+n=0$ items)

$\Rightarrow$ 工作中, 保留 3阶 Magnus expansion.

那么 $[H_i, [H_m, H_n]]$ 有 $i+m+n=0$

For example: $[H_{-1}, [H_{-3}, H_2]]$ $\rightarrow$ guess $g(\omega t)\, X_1 X_2 + g(-3\omega t)(X_2 X_3 + Y_2 Y_3) + g(2\omega t)\, X_3 X_4$

$XZZX \propto [X_1 X_2, [X_2 X_3 + Y_2 Y_3, X_3 X_4]]$

Determining the driving fields

- The target operator can be expressed with two-spin operators:

$$\hat{X}_1\hat{Z}_2\hat{Z}_3\hat{X}_4 \propto \left[\hat{X}_1\hat{X}_2, \left[\hat{X}_2\hat{X}_3 + \hat{Y}_2\hat{Y}_3, \hat{X}_3\hat{X}_4\right]\right]$$

- Guess the periodic driven function with Magnus expansion.
- Optimize the driving fields with single-plaquette simulation.

Lie closure

Lie closure

- The Lie closure can be computed by the Pennylane library:

```
ops = [...] "All operators in driven Hamitonian"
dla = qml.lie_closure(ops)
print(dla)
```

- Lie closure: all possible nested commutators of the generators $\hat{G}_j$ in driven Hamitonian.

```
>>> print(dla)
[X(0) @ X(1) + Y(0) @ Y(1), X(1) @ X(2) + Y(1) @ Y(2), X(2) @ X(3) + Y(2) @ Y(3), X(0), X(3), X(0) @ Z(1) @ Y(2) + -1.0 *
(Y(0) @ Z(1) @ X(2)), -1.0 * (Z(0) @ Y(1)), X(1) @ Z(2) @ Y(3) + -1.0 * (Y(1) @ Z(2) @ X(3)), -1.0 * (Y(2) @ Z(3)), (
    -1.0 * (X(0) @ Z(1) @ Z(2) @ X(3))
  + -1.0 * (Y(0) @ Z(1) @ Z(2) @ Y(3))
), Z(0) @ Z(1) @ X(2), -1.0 * (Y(0) @ Y(1)), -1.0 * (X(1) @ Z(2) @ Z(3)), -1.0 * (Y(2) @ Y(3)), Z(0) @ Z(1) @ Z(2) @ Y(3),
 Y(0) @ Z(1) @ Z(2) @ Z(3), Y(0) @ Z(1) @ X(2), -1.0 * (X(1) @ Z(2) @ Y(3)), Y(0) @ Z(1) @ Z(2) @ Y(3), -1.0 * (Z(0) @ Z(1
) @ Z(2) @ Z(3)), -1.0 * (X(1) @ X(2))]
```

Lie closure

Lie closure

- The table in page3 can be calculated with Dynamical Lie Algebra.(https://pennylane.ai/qml/demos/tutorial_liealgebra)

Comutators1:

```
>>> print(lie_closure_1)
[0.0 * I, 31.200000000000003j * Z(0) @ Y(1), 7j * Y(2) @ Z(3), 6.760000000000001j * X(0) @ Z(1) @ Y(2), -6.760000000000001j * Y
(0) @ Z(1) @ X(2), 1.8199999999999998j * X(1) @ Z(2) @ Y(3), -1.8199999999999998j * Y(1) @ Z(2) @ X(3)]
```

commutator2:

```
>>> print(lie_closure_2)
[(81.12+0j) * Z(0) @ Z(1) @ X(2), (-18.2+0j) * X(1) @ Z(2) @ Z(3), (2.3659999999999997+0j) * Z(1) @ Z(2) @ X(3) @ X(0), (2.3659
999999999997+0j) * Z(1) @ Z(2) @ Y(3) @ Y(0)]
```

commutator3:

Lie closure

```
>>> print(lie_closure_3)
[28.391999999999996j * Z(1) @ Z(2) @ Y(3) @ Z(0), 23.659999999999997j * Z(1) @ Z(2) @ Z(3) @ Y(0)]
>>>
```

commutator4:

```
>>> print(lie_closure_4)
[(-283.91999999999996+0j) * Z(1) @ Z(2) @ Z(3) @ Z(0)]
>>>
```

Lie- Algebra decoupling theorem

Lie- algebra decoupling theorem

- Originally outlined by Wei and Norman[1].
- the non-Gaussian character of quantum optomechanical systems evolving under the fully nonlinear optomechanical Hamiltonian.[2], [3], [4], [5], [6], [7]
- Floquet engineering of Lie algebraic quantum systems[8]
- Floquet engineering with high frequency approximation.[9], [10], [11]
- High-frequency expansions for time-periodic Lindblad generators[12]
- Lie-algebraic classical simulations for quantum computing[13]

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